SUPPORTING ADULT GUIDANCE

Substituting positive, negative and fractional values

Formulae and Rearranging · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson develops accurate substitution of positive, negative and fractional values into expressions and formulae. Learners predict and then watch a substitution machine show each stage in the order of operations, explore the difference between -3^2 and $(-3)^2$, find slips in worked substitutions, and substitute into real formulae before interpreting the answer in context.

Driving Question: *What happens when the numbers get awkward?*

The mathematics, in plain terms

Substitution replaces each variable with its value; writing each value in brackets preserves its sign and makes the order of operations unambiguous. Powers bind more tightly than a leading minus sign, so $-3^2 = -(3^2) = -9$, while $(-3)^2 = 9$. The product of two negatives is positive; subtracting a negative is equivalent to adding its absolute value.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A negative value with a square

Work out $3x^2 - 2x$ when $x = -2$.

1. Substitute, in brackets.	$3 \times (-2)^2 - 2 \times (-2)$
2. Powers first: $(-2)^2 = 4$.	$3 \times 4 - 2 \times (-2)$
3. Then multiplication.	$12 - (-4)$
4. Subtracting a negative is adding.	$12 + 4 = 16$

$3x^2 - 2x = 16$ when $x = -2$.

Example 2 — A real formula with a negative value

Use F = 1.8C + 32 to convert –10 °C to °F.

1. Substitute C = –10, in brackets.	F = 1.8 × (–10) + 32
2. Multiplication first.	F = –18 + 32
3. Then addition.	F = 14
4. Does it make sense? Freezing is 32 °F.	14 °F is below freezing ✓

–10 °C is 14 °F.

Predict, then run

In the substitution machine, learners type a prediction before the stages appear. A wrong prediction is useful: the machine names the likely slip (a lost bracket, a negative squared as negative, subtracting a negative). Encourage learners to compare their own working with each stage, rather than just the final answer.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Bracket or no bracket? Predict -3^2, $(-3)^2$, $-x^2$ and $(-x)^2$, then see why.	~7 min
Movement	The substitution machine: predict, then watch each stage in order.	~15 min
Reflection	Find the slip in worked substitutions — or decide there is none.	~9 min
Creativity	Real formulae with negatives and fractions, then interpret the answer.	~9 min
Rest	Brackets as a pair of hands carrying a negative number safely.	~2 min
Nutrition	Intellectual: care with detail where mistakes really happen.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Losing the sign: writing 4×-2 as $4 - 2$, or $(-3)^2$ as -9.
Insist on brackets for every substituted value, and ask: 'What is negative times negative?'

Treating $-x^2$ as $(-x)^2$.
Ask which part the square is attached to, and compare both with $x = -4$.

$10 - (-6)$ taken as 4.
Use a number line or a temperature context: removing a debt of 6 increases the total by 6.

Adding before multiplying or dividing ($x \div 4 + 1$ as $x \div 5$).
Ask the learner to say the order of operations aloud before starting.

Rounding fractions away (3½ hours taken as 3).
Ask the learner to write the fraction as a decimal or keep it as a fraction throughout.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: calculating with negative numbers, and the notation from Lesson 1. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says $2x^2 = -18$ when $x = -3$. The aim is a question that lets them find the slip themselves.

- "Can you write it again with -3 in brackets?"
- "What is $(-3) \times (-3)$?"

- “Which part of $2x^2$ is squared?”
- “Could a square ever be negative?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why insist on brackets when the number is positive?

The habit matters most when it is needed, so practising it every time makes it automatic for negatives and fractions.

My calculator gives –9 for -3^2. Is it wrong?

No. It follows the convention that the power is applied before the minus sign. To square –3, type $(-3)^2$.

Should answers be fractions or decimals?

Either is accepted throughout the lesson: 7.5, 15/2 and 7½ are all correct.

What comes next?

Lesson 3 writes formulae from words and diagrams, and then uses them.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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